

A Proof of Conjecture 00000000953: No Real ETF(17, 8), and a Real ETF(16, 6)

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Abstract

Conjecture 00000000953 asserts that a real equiangular tight frame of 17 vectors in $\mathbb{R}^8$ does not exist, while one of 16 vectors in $\mathbb{R}^6$ does. We prove both parts. For the first, the Gram matrix of such a frame would force the integrality relation $9m^2 = 2$. For the second, we give the classical example of 16 sign vectors in $\mathbb{R}^6$. Both parts are formalized in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000000953 reads:

Definition: ETFs (equiangular tight frames). Conjecture: The real ETF(17, 8) does not exist while ETF(16, 6) does (boundary entries of the small-parameter ETF table, computationally verifiable).

2 Reading of the statement

Definition 1. A *real equiangular tight frame* of N vectors in $\mathbb{R}^d$ consists of unit vectors $f_1, \dots, f_N \in \mathbb{R}^d$ with

- (equiangular) $|\langle f_i, f_j \rangle| = \alpha$ for all $i \neq j$, for some constant α ;
- (tight) $\sum_i \langle f_i, x \rangle f_i = Ax$ for all $x \in \mathbb{R}^d$, for some constant A .

ETF(N, d) denotes N vectors in $\mathbb{R}^d$. This is the only consistent reading: a frame of N vectors in $\mathbb{R}^d$ needs $N \geq d$, so ETF(16, 6), which the statement says exists, must mean 16 vectors in $\mathbb{R}^6$. The conjecture is therefore the conjunction: no real ETF of 17 vectors in $\mathbb{R}^8$ exists, and a real ETF of 16 vectors in $\mathbb{R}^6$ exists.

3 Result

Theorem 2. *There is no real equiangular tight frame of 17 vectors in $\mathbb{R}^8$, and there is one of 16 vectors in $\mathbb{R}^6$. Hence Conjecture 00000000953 holds.*

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4 Proof

4.1 Nonexistence of ETF(17, 8)

Let $f_1, \dots, f_{17}$ be a real ETF in $\mathbb{R}^8$ and $G_{ij} = \langle f_i, f_j \rangle$ its Gram matrix.

Lemma 3. $G^2 = AG$, and $A \cdot d = N$, i.e. $A = 17/8$.

Proof. $(G^2)_{ij} = \sum_k \langle f_i, f_k \rangle \langle f_k, f_j \rangle = \langle f_i, \sum_k \langle f_k, f_j \rangle f_k \rangle = A G_{ij}$. For the second claim, take the trace of the frame operator in an orthonormal basis (e_k) : by Parseval, $\sum_k \langle e_k, \sum_i \langle f_i, e_k \rangle f_i \rangle = \sum_i \|f_i\|^2 = N$, while it also equals $\sum_k \langle e_k, A e_k \rangle = Ad$. $\square$

Since $G_{ii} = 1$ and $G_{ik}^2 = \alpha^2$ for $k \neq i$, the diagonal of $G^2 = AG$ reads $1 + 16\alpha^2 = A = 17/8$, so $\alpha^2 = 9/128$. Now look at the entry $(1, 2)$:

$$(G^2)_{12} = G_{11}G_{12} + G_{12}G_{22} + \sum_{k \neq 1,2} G_{1k}G_{k2} = 2G_{12} + \alpha^2 m,$$

where each product $G_{1k}G_{k2}$ has square α^4 , hence equals $\pm\alpha^2$, and $m \in \mathbb{Z}$ is the sum of these signs. With $(G^2)_{12} = \frac{17}{8}G_{12}$ this gives $\frac{1}{8}G_{12} = \frac{9}{128}m$, i.e. $G_{12} = \frac{9}{16}m$. Squaring, $\frac{9}{128} = \alpha^2 = \frac{81}{256}m^2$, so $9m^2 = 2$, which has no integer solution. Hence no such frame exists.

4.2 Existence of ETF(16, 6)

Let $v_1, \dots, v_{16}$ be the vectors $(1, x_2, \dots, x_6) \in \{\pm 1\}^6$ with $x_2 x_3 x_4 x_5 x_6 = 1$, and $f_i = v_i / \sqrt{6}$. Each f_i is a unit vector. For $i \neq j$, v_i and v_j differ in an even, nonzero number of coordinates other than the first, i.e. in 2 or 4 places, so $\langle v_i, v_j \rangle = 6 - 2 \cdot \#\{\text{differences}\} = \pm 2$ and $|\langle f_i, f_j \rangle| = 1/3$. Finally $\sum_i v_i v_i^\top = 16I$: the diagonal entries are 16, and an off-diagonal entry $\sum_i v_{ik} v_{il}$ vanishes because the set is invariant under flipping two of the coordinates $x_2, \dots, x_6$, which changes the sign of $v_{ik} v_{il}$. Hence $\sum_i \langle f_i, x \rangle f_i = \frac{16}{6}x$, a tight frame with $A = 8/3 = N/d$. These are the 16 equiangular lines in $\mathbb{R}^6$ [1].

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states the conjecture and proves it against Mathlib, in `EuclideanSpace ℝ (Fin d)`.

- `IsETF N d f, ETFExists N d` — the definition above: unit norms, $\exists \alpha$ with $|\langle f_i, f_j \rangle| = \alpha$ for $i \neq j$, and $\exists A$ with $\sum_i \langle f_i, x \rangle f_i = Ax$ for all x ; `ConjectureHolds =  $\neg$ ETFExists 17 8  $\wedge$  ETFExists 16 6`.
- `gram_sq, trace_eq` — Lemma 3; the trace uses Parseval's identity in the standard orthonormal basis of $\mathbb{R}^d$ (Mathlib's `sum_inner_mul_inner`).
- `not_etf_17_8` — the nonexistence argument.
- `v, v_gram, v_norm, v_frame` — the sign vectors and their integer identities, checked by `decide`; `F, norm_F, abs_inner_F, tight_F, etf_16_6` — the frame.
- `conjecture_00000000953` — `ConjectureHolds`.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000000953
```

reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

The nonexistence is an instance of the integrality conditions of Sustik, Tropp, Dhillon and Heath [2]. For a real ETF of $N \neq 2d$ vectors in $\mathbb{R}^d$, the numbers $\sqrt{d(N-1)/(N-d)}$ and $\sqrt{(N-d)(N-1)/d}$ must be odd integers. For $(N, d) = (17, 8)$ the first is $\sqrt{128/9}$. The argument above is a direct, self-contained version of this obstruction.

References

- [1] P. W. H. Lemmens and J. J. Seidel, *Equiangular lines*, J. Algebra 24 (1973), 494–512.
- [2] M. A. Sustik, J. A. Tropp, I. S. Dhillon and R. W. Heath, *On the existence of equiangular tight frames*, Linear Algebra Appl. 426 (2007), 619–635.